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Adaptation to a novel host and performance trade-off in host-generalist and host-specific insect ectoparasites

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Abstract We investigated the performance trade-offs of fleas (Siphonaptera) while adapting to a novel host using two host generalists (*Xenopsylla conformis* and *Xenopsylla ramses*) and one host specialist (*Parapulex chephrenis*) maintained on their principal hosts (*Meriones crassus* for *Xenopsylla* and *Acomys cahirinus* for *P. chephrenis*). We asked whether, over generations, (i) a host generalist may become a specialist by evolving the ability to exploit a novel host and losing the ability to exploit an original host and (ii) a host specialist can become a generalist by evolving the ability to exploit a novel host without losing the ability to exploit an original host. We established an experimental line of each species on a novel host (*Acomys russatus* for *Xenopsylla* and *M. crassus* for *P. chephrenis*) and maintained this line on this host during 23 generations. We compared reproductive performance of progenitors of each line and their descendants when they exploited either original or novel host in terms of egg number and size, hatching success, offspring production, and offspring size. We found changes in performance over generations in female offspring size only. *Xenopsylla conformis* demonstrated a tendency to become a host specialist (increased performance on the novel host with a concomitant decreased performance on the original host), whereas *P. chephrenis* demonstrated a tendency to become a host generalist (increased performance on the novel host without a concomitant decreased performance on the original host). We conclude that the probability of generalist to specialist transition, and vice versa, is context-dependent and varies between species.

Key words experimental evolution; fleas; generalist; performance; rodents; specialist

Introduction

Understanding the evolution of ecological specialization is essential for elucidating the drivers of adaptive radiation and assessing risks of extinction (Futuyma & Moreno, 1988; Thompson, 1994; Poisot *et al.*, 2011). Despite significant research (Levins, 1968; Agrawal, 2000; Bolnick *et al.*, 2003; Legros & Koella, 2010), many aspects of this phenomenon are still poorly known

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(Agrawal, 2020) with many studies reporting contradictory results (e.g., Bennet & Lensky, 2007). For example, the possibility of the evolutionary transition from specialist to generalist and the occurrence of a trade-off between the abilities to adapt to a new resource and to exploit the original resource are still debated (Morse & Farrell, 2005; Agosta & Klemens, 2009; Sriswasdi *et al.*, 2017).

Since Mayr (1963), the commonly accepted paradigm was that specialization leads into blind alleys, and that an evolutionary transition from generalist to specialist is much more likely than the opposite (Futuyma & Moreno, 1988; Jaenike, 1990). Yet, recent research on phytophagous insects suggest that this is not always the case (e.g., Morse & Farrell, 2005; Nosil, 2005) and the paradigm has been challenged (Thompson, 1994; Agosta *et al.*, 2010). Evolution of specialization in parasitic arthropods has received much less attention, even though this subject is of the utmost importance for understanding their evolution because the level of specialization (defined for parasites as host specificity) is a fundamental property of a parasite species affecting a variety of their traits (Poulin *et al.*, 2011). Furthermore, patterns found in phytophagous arthropods and patterns found in parasitic arthropods may differ due to the extremely tight and intimate relationships of the latter with their resources (i.e., their hosts) (Combes, 2001). Although evolution of host specificity has been considered in comparative analyses (see review in Poulin *et al.*, 2011), only a few studies have addressed it experimentally (e.g., Giorgi *et al.*, 2004; Dick *et al.*, 2009; Arbiv *et al.*, 2012; Dietrich *et al.*, 2014). Nevertheless, phylogenetic studies on arthropod parasites also indicated that high host specificity is not a proverbial “one-way ticket,” and that evolution of host specificity is reversible and nondirectional (Poulin *et al.*, 2006; Klimov & OConnor, 2013). However, the transition of insect parasites from specialist to generalist or vice versa has rarely been specifically tested experimentally (Arbiv *et al.*, 2012; Dietrich *et al.*, 2014). Previous studies used only one or a few parasite generations, so that the results cannot provide clear-cut answers about the probability of a specialist-generalist (or vice versa) transition.

The probability of evolutionary transition of a parasite from generalist to specialist or vice versa can be inferred from the pattern of adaptation to a new resource and whether it is accompanied by the decreased ability to exploit an original resource. In other words, the direction of transition between specialist and generalist may depend on the cost of host specificity, which is manifested as a trade-off between the abilities to extract resources from different hosts. For example, a generalist adapting to a novel host and experiencing concomitant loss of the ability to exploit an original host is on its evolutionary way to

become a specialist. Likewise, a specialist adapting to a novel host and retaining the ability to exploit an original host is on its evolutionary way to become a generalist.

Although numerous studies of phytophagous arthropods and microorganisms addressed the effects of adaptations to one host on performance on alternative hosts (e.g., Fry, 1996; Ebert, 1998; Agrawal, 2000; Duffy *et al.*, 2006; Magalhães *et al.*, 2009; Schulte *et al.*, 2011; Forister *et al.*, 2012), no negative correlation between relative performance on different hosts was found in many of them (e.g., Magalhães *et al.*, 2009, but see Via, 1991). However, the pattern for parasites may differ from that of, for example, phytophagous arthropods because a parasite’s adaptations to a host include not only the development of tools to utilize the food resources provided by that host but also the tools to overcome immediate behavioral or immunological host defense efforts.

In general, our knowledge regarding the costs of high host specificity in parasites in terms of performance trade-offs in different hosts is limited. In fact, trade-offs in parasite performance between alternative hosts have been found in some (Dobson & Owen, 1977; Arbiv *et al.*, 2012) but not in other studies (Jaenike & Dombeck, 1998; Stavtseva *et al.*, 2021). One of the reasons for these inconsistencies could be the context-dependence of performance trade-offs (i.e., variation of trade-offs among parasite–host associations) associated with the original level of host specificity. For example, theoretical models indicate that trade-offs arise only in parasite species “familiar” with (i.e., adapted to) more than one host (Joshi & Thompson, 1995). If so, then a performance trade-off will be found mainly in generalist, rather than specialist species, because a specialist is, by definition, not adapted to more than one host. This may result in a generalist evolving to be a specialist rather than vice versa. However, adaptations to a new host may evolve without compromising performance on the original host due to conservatism in traits related to host exploitation (i.e., when a lineage preserves plasticity in host use) (Agosta & Klemens, 2009; Agosta *et al.*, 2010). In other words, a parasite can be “preadapted” to use a variety of hosts, including novel ones. “Preadaptations” to use several hosts are more likely to occur in generalists than specialists as has been indirectly supported by experiments with fleas (Khokhlova *et al.*, 2012a). From this perspective, performance trade-offs would be more likely in an originally host-specific parasite than in an originally host-opportunistic parasite. This may result in a specialist evolving to be a generalist rather than vice versa. However, the effect of the original level of specialization on the evolution of host specificity and accompanying performance trade-offs has never been experimentally tested.

Here, we investigated the occurrence of performance trade-offs while adapting to a novel host using fleas (Siphonaptera) as a model taxon. Fleas are holometabolous insect ectoparasites most abundant and diverse on small burrowing mammals with mainly non-parasitic preimagos developing off the host and imagoes alternating periods on and off the host. This makes fleas a very convenient model parasite in laboratory studies because their infestation of living hosts can be easily manipulated (Krasnov, 2008). We used long-term laboratory colonies of two host-generalist and one host-specialist flea species maintained on their preferred hosts. We established an experimental line of each flea species on a novel host and maintained this line on this host during 23 generations. We asked (i) whether a host generalist, over generations, may become a host specialist by evolving the ability to exploit a novel host and concomitantly losing the ability to exploit an original host and (ii) whether a host specialist, over generations, can become a host generalist by evolving the ability to exploit a novel host without losing the ability to exploit an original host. To answer these questions, we compared reproductive performance of progenitors of each line and their descendants of the 23rd generation when they exploited their original and a novel host. Reproductive performance of fleas was assessed in terms of quantity and quality of their offspring (see details below).

Materials and methods

Maintenance of fleas and hosts

We used three flea species (*Xenopsylla conformis* (Wagner, 1903), *X. ramesis* (Rothschild, 1904), *Parapulex chephrenis* (Rothschild, 1903)) and three rodent species (Sundevall's jird *Meriones crassus* Sundevall, 1842; the Egyptian spiny mouse *Acomys cahirinus* É. Geoffroy, 1803 and golden spiny mouse *Acomys russatus* Wagner, 1840) from our laboratory colonies started in 1998–1999 from field-collected individuals. Both *Xenopsylla* species are host generalists that exploit multiple (seven in the Negev desert) gerbil species belonging to five genera/subgenera being most abundant and prevalent on *M. crassus* and only occasionally recorded on spiny mice of the genus *Acomys* I. Geoffroy, 1838 (Krasnov *et al.*, 1999). Two *Acomys* species are specific hosts of *P. chephrenis*, which has rarely been found on any other rodent species (Krasnov *et al.*, 1999). Our flea colonies were bred on their main hosts (*X. conformis* and *X. ramesis* on *M. crassus* and *P. chephrenis* on *A. cahirinus*). To maintain genetic diversity of the colonies, we added 150–

200 field-collected fleas of each species to a respectively colony yearly.

Details on colony maintenance have been previously published (e.g., Krasnov *et al.*, 2001, 2003a; Khokhlova *et al.*, 2009, 2012b). In brief, fleas were maintained on rodents kept individually in plastic cages. Every two weeks, all substrate and bedding material from the rodent's nest box and pan were collected into plastic boxes with perforated lids and transferred to an incubator (FOC225E, Velp Scientifica srl, Milano, Italy; 25°C air temperature and 75% relative humidity) where the fleas developed. Newly emerged fleas collected from these incubation boxes were used for infestation of a new host.

Rodents were maintained in plastic cages (60 × 50 × 40 cm) at 25°C air temperature with a photoperiod of 12 : 12 (L : D) hours, and with sawdust as bedding material and offered millet seeds and fresh alfalfa leaves *ad libitum* daily as well as commercial cat chow once a week (for *Acomys*). Every year or two, we added about 10–15 field-captured rodents of each species to our colonies to maintain genetic diversity. Initially, a male and a female rodent born from different parents or grandparents were placed in each cage to produce offspring subsequently used in experiments. Resulting young were transferred to a new cage at one month of age. In this study, we used adult males (6–8 months old) taken from a respective colony and maintained individually or in pairs for at least 2 weeks prior to experiments.

Experimental design

For each flea species maintained on its original (= principal) host we established a new laboratory line by maintaining and rearing them on a novel host (both *Xenopsylla* on *A. russatus* and *P. chephrenis* on *M. crassus*). Two hundred newly emerged fleas of each species were randomly selected from their respective colony and then released into plastic cages, each with two or three individuals of the alternative host. These fleas usually start oviposition 2 days after being released into a host cage (personal observations). Consequently, we collected all substrate material from each cage on day 3 and also collected fleas from rodent's bodies using a toothbrush over a white plastic pan. Then, we placed the cage substrate material, as well as parent fleas that survived, into a plastic box and kept it at 25°C and 90% relative humidity in an incubator (FOC225E, Velp Scientifica srl, Milano, Italy). Minimum duration of the egg-to-imago development in *X. conformis* and *X. ramesis* is 24 days and in *P. chephrenis* 34 days (Khokhlova *et al.*, 2014; van der Mescht *et al.*, 2019). We collected new imagoes of *Xenopsylla*

from incubated boxes on day 32 and new imagoes of *P. chephrenis* on day 42, so that all of them would belong to the same generation and placed them on new host individuals. These procedures were repeated until the 23rd generation of each line. We measured reproductive performance (see below) in fleas of parent generation (= progenitors) and fleas from 23rd generation of each experimental line (= descendants) after they were forced to feed on either their original host or the novel host. The earliest common ancestors of rodents used as hosts for subsequent generations of each flea line were their great grandparents.

Measurements of reproductive performance

To measure reproductive performance of a progenitor generation, we randomly selected 200 fleas (130 females and 70 males) of each species and placed them into a plastic cage with sand substrate and two or three rodents of either an original or a novel host species. After 48 hours, 35–70 female fleas (depending on the number of surviving females) were randomly selected from each cage (i.e., from the substrate and rodent's body), examined under a light microscope without dissection to verify engorgement and randomly distributed into seven Petri dishes with 5–10 fleas per dish. These Petri dishes were checked for oviposition the next morning using a digital microscope camera, Moticam 2000 with Motic Images Plus 2.0ML program (Motic, Speed Fair Cp., Ltd, Causeway Bay, Hong Kong). We counted the total number of eggs produced by each group of females (i.e., females kept in the same Petri dish) and the number of dead females. A photograph was taken of every egg and maximal length and width of an egg were measured on-screen up to nearest 0.01 mm with 40 $\times$ magnification and calibration using an object-micrometer.

Each Petri dish with eggs was then filled with a 1 mm layer of sand and larval food medium (94% dry bovine blood, 5% millet flour, and 1% ground excrement of the host on which fleas fed) and maintained at 25°C air temperature and 90% RH. Starting from the 18th day (for *Xenopsylla*) or 28th day (for *P. chephrenis*) after eggs were laid (i.e., approximately a week less than minimal egg-to-imago duration), we checked each Petri dish daily either until all adults emerged or for 60 consecutive days. We recorded the day of emergence of each new imago and counted new imagoes produced by each group of parent fleas. After emergence, each adult was placed in an individual Eppendorf vial and left in the incubator at the same air temperature and RH. Vials with new imagoes were checked daily until all adults died.

After the death of each imago, we identified its sex by examining its genitalia under light microscopy and estimated its body size by measuring the maximal length of its right and left hind femurs. The length of a femur was shown to be a reliable indicator of body size in fleas because of strong correlation between these traits (Krasnov et al., 2003b). Femur length was measured on a screen similarly to measurement of eggs (see above).

The same measurements were carried out for descendant (23rd generation) of each species except that the number of descendant fleas released into each rodent's cage varied from 50 to 200 (depending on the number of new imagoes available). In total, 427 progenitor females produced 450 eggs and 391 new imagoes, whereas 613 descendant fleas produced 717 eggs and 501 new imagoes.

Data analyses

We assessed reproductive performance via mean number of eggs produced per female for a group of fleas (i.e., females kept in the same Petri dish), volume of an egg, mean number of new imagoes produced per female for a group of fleas and size of new imago (using femur length averages between right and left femur as a proxy and separately for males and females due to strong female-biased sexual size dimorphism in fleas; Krasnov, 2008). Mean number of eggs and new imagoes per female was calculated by dividing the total number of eggs by the total number of females in a group of fleas being alive on the first day of oviposition. In addition, we calculated hatching success as mean number of eggs developed into new imagoes per group of fleas. Egg volume was used as a proxy for egg size and was calculated as $V = 1/6\pi \times W^2 \times L$, where V is egg volume, W is maximal egg width and L is maximal egg length (Berrigan, 1991).

Then, we analyzed the effects of host identity (the original vs. the novel host) and generation (progenitors of the experimental line vs. their descendants) on reproductive performance in terms of quantity and quality of the offspring. Offspring quantity was estimated by egg and new imago production (the mean number of eggs or new imagoes produced per female flea, respectively), whereas offspring quality was estimated by egg size, hatching success and the offspring size.

The effects of host identity and generation as well as the interaction between them (=independent variables) on the mean number of eggs and new imagoes produced per female flea and hatching success (=dependent variables) were analyzed using generalized linear models (GLM) with the normal distribution and identity link

function using the package “stats” of the R 4.0.3 statistical environment (R Core Team, 2020) separately for each flea species. The effects of host identity, generation, and their interaction on egg volume and offspring size were analyzed using linear mixed-effects models (LMEs) implemented in “lme4” (Bates *et al.*, 2015) and “lmerTest” (Kuznetsova *et al.*, 2017) packages in R. In addition, we included the number of eggs produced per female flea as another independent variable in the analyses of the above-mentioned effects on egg size to account for possible trade-off between the number and size of eggs. We also included total number of new imagoes produced by a group of fleas and the duration of development of each new imago in the analyses of new imago size because of possible relationships between larval density (number of eggs laid by all females in a group as a proxy), larval development and the size of a new imago (Khokhlova *et al.*, 2010; Kiefer *et al.*, 2016). To account for variation between replicates, we included a code number of a group of female fleas from the same Petri dish as random effect in each mixed-effects model. Marginal means for LMEs were estimated using command “emmeans” from the R package “emmeans” (Length, 2020). We used Tukey’s HSD tests for multiple comparisons with Bonferroni adjustment of alpha level implemented in the package “emmeans.” Performance of each mixed-effects model was assessed via the marginal and the conditional R^2 (Nakagawa *et al.*, 2017). The marginal R^2 indicates proportion of the model’s variance explained by the fixed effects only, whereas the conditional R^2 indicates proportion of variance explained by the full model while taking into account fixed and random effects. The marginal and the conditional R^2 were calculated using the R package “performance” (Lüdtke *et al.*, 2020). The distribution of response variables did not significantly deviate from normality (Kolmogorov–Smirnov tests; $d = 0.06–0.09$, $P > 0.20$ for all), so no transformations were applied. We tested normality of residuals by visually inspecting plots of sample quantiles against theoretical quantiles produced using function “qqnorm” of the R package “stats.” No deviation from normality was found in either model.

Results

In all three flea species, egg and new imago production was significantly affected by host identity (Table 1), whereas generation influenced the production of new imagoes only in *X. ramesis* (Table 1). In both progenitors and descendants of experimental lines, egg and offspring production was consistently higher when fleas ex-

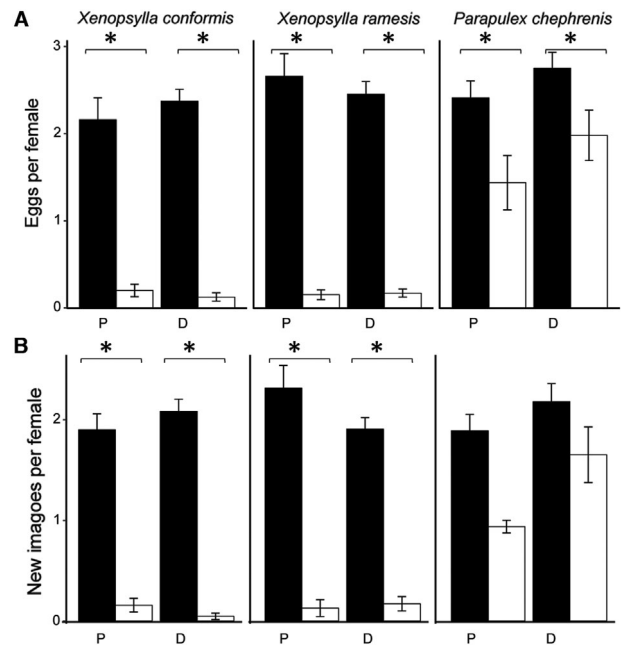


Fig 1 Mean ($\pm$ SE) number of eggs during one day of oviposition (A) and new imagoes (B) produced by female *Xenopsylla conformis*, *Xenopsylla ramesis*, and *Parapulex chephrenis* after 48-hour stay on either an original host (black columns) or a novel host (white columns). The original host is the host, which fleas of a given species infest naturally. Laboratory colonies of fleas were bred on their original hosts (*X. conformis* and *X. ramesis* on *Meriones crassus*, *P. chephrenis* on *Acomys cahirinus*) and then maintained on the novel hosts (*X. conformis* and *X. ramesis* on *Acomys russatus*, *P. chephrenis* on *M. crassus*). Reproductive performance after feeding on either the original or the novel host was estimated for progenitors of each line (parent generation; P) and their descendants of 23rd generation (D). Asterisks denote significant differences. Note that the difference between offspring production between progenitor and descendant *P. chephrenis* became nonsignificant after Bonferroni adjustment of alpha level.

ploited the original host (Tukey’s tests within a generation between hosts; t -ratio = 3.20–14.88, $P < 0.01$ for all; Fig. 1) except for offspring production in *P. chephrenis* (t -ratio = 2.04 and t -ratio = 1.84 for progenitors and descendants, respectively; $P > 0.20$ for both). On the contrary, there was no difference in either egg or offspring production between progenitors and descendants fed on the same host species in either flea species (Tukey’s tests between generations within host; t -ratio = -1.59 to 2.20 , $P > 0.13$ for all; Fig. 1).

Hatching success was affected by host identity in *X. ramesis* and interaction between host identity and generation in *X. conformis* (Table 1), whereas in *P. chephrenis* the number of new imagoes hatched per egg did not dif-

Table 1 Summary of the generalized linear models of the effect of host identity (HI) and generation (GEN) and their interaction on reproductive performance of *Xenopsylla conformis* (Xc), *Xenopsylla ramesis* (Xr), and *Parapulex chephrenis* (Pc) estimated via the number of eggs (NE) laid per female, number of new imagoes (NI) produced per female and hatchings success (HS; number of new imagoes hatched per egg). Fleas were bred on their original hosts (*X. conformis* and *X. ramesis* on *Meriones crassus*, *P. chephrenis* on *Acomys cahirinus*) and then maintained on the novel hosts (*X. conformis* and *X. ramesis* on *Acomys russatus*, *P. chephrenis* on *M. crassus*). Reproductive performance after feeding on either the original or the novel host was estimated for progenitors of each line (parent generation) and their descendants of 23rd generation (independent variable GEN). RV, response variable; EV, explanatory variable. Reference levels of explanatory variables were the original host for HI and parent generation for GEN. *F*-statistics and the respective *P*-value are related to the entire model. Significant *P*-values are in bold.

Flea	RV	EV	Coefficient ± SE	<i>t</i>	<i>P</i>	<i>F</i>	<i>P</i>
Xc	NE	HI	−1.96 ± 0.16	−8.63	<0.001	70.88	<0.001
		GEN	0.21 ± 0.20	1.05	0.30		
		HI × GEN	−0.29 ± 0.29	−0.96	0.34		
	NI	HI	−1.74 ± 0.17	−9.88	<0.001	94.76	<0.001
		GEN	0.18 ± 0.15	1.16	0.26		
		HI × GEN	−0.29 ± 0.23	−1.25	0.22		
	HS	HI	−0.20 ± 0.15	−1.32	0.20	4.53	0.01
		GEN	−0.02 ± 0.12	−0.18	0.85		
		HI × GEN	−0.30 ± 0.11	−2.43	0.02		
Xr	NE	HI	−2.51 ± 0.20	−12.51	<0.001	130.50	<0.001
		GEN	−0.20 ± 0.19	−1.09	0.28		
		HI × GEN	0.22 ± 0.25	0.89	0.38		
	NI	HI	−2.18 ± 0.22	−10.09	<0.001	73.58	<0.001
		GEN	−0.41 ± 0.18	−2.20	0.03		
		HI × GEN	0.45 ± 0.27	1.64	0.11		
	HS	HI	−0.51 ± 0.16	−3.15	0.003	7.50	<0.001
		GEN	−0.09 ± 0.13	−0.65	0.52		
		HI × GEN	0.09 ± 0.20	0.44	0.66		
Pc	NE	HI	−0.97±0.50	−2.23	0.04	3.57	0.02
		GEN	1.04±1.04	1.22	0.23		
		HI × GEN	0.20±0.59	0.34	0.74		
	NI	HI	−0.95 ± 0.43	−2.30	0.04	3.04	0.04
		GEN	0.29 ± 0.26	1.12	0.27		
		HI × GEN	0.45 ± 0.54	0.77	0.44		
	HS	HI	−0.10 ± 0.08	−1.25	0.22	1.24	0.31
		GEN	0.01 ± 0.05	0.20	0.84		
		HI × GEN	0.16 ± 0.10	1.56	0.13		

fer neither within generation between hosts nor between generations fed on the same host (Tukey's tests; *t*-ratio = −0.95 to 1.88, *P* > 0.20 for all). In descendant, but

not progenitor, *X. conformis*, more eggs survived until hatching if fleas fed on the original than on the novel host (Tukey's test, *t*-ratio = 3.38, *P* = 0.01) (Fig. 2). In

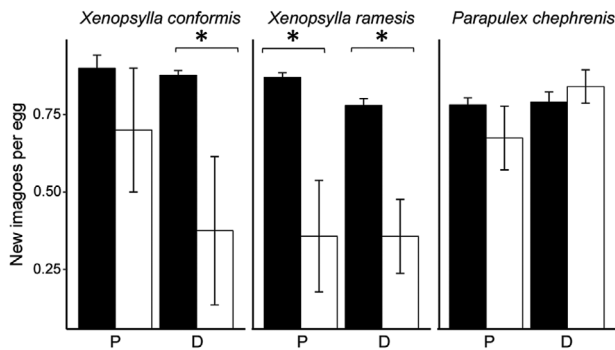


Fig 2 Mean ($\pm$ SE) hatching success (number of new imagoes hatched per egg) of eggs produced by female *Xenopsylla conformis*, *Xenopsylla ramesis*, and *Parapulex chephrenis* after 48-hour stay on either an original host (black columns) or a novel host (white columns). The original host is the host, which fleas of a given species infest naturally. Laboratory colonies of fleas were bred on their original hosts (*X. conformis* and *X. ramesis* on *Meriones crassus*, *P. chephrenis* on *Acomys cahirinus*) and then maintained on the novel hosts (*X. conformis* and *X. ramesis* on *Acomys russatus*, *P. chephrenis* on *M. crassus*). Hatching success of eggs produced by fleas after feeding on either the original or the novel host was estimated for progenitors of each line (parent generation; P) and their descendants of 23rd generation (D). Asterisks denote significant differences.

both progenitor and descendant *X. ramesis*, hatching success was higher in fleas exploiting the original host as compared with the novel host (Tukey's tests; t -ratio = 0.16 and t -ratio = 3.38, $P < 0.01$ for both; Fig. 2).

No effect of either host species or generation was found for egg size in all three species (Table 2). The same was true for the effect of clutch size (i.e., number of eggs produced per female) except for *P. chephrenis* in which larger clutches were composed of larger eggs (Table 2). In general, both marginal and conditional R^2 in models for all three flea species were extremely low (Table 2).

The effects of host identity, generation, and interaction between them on the size of female offspring differed between the three flea species. Interaction between a host's identity and generation was significant in *X. conformis* and *P. chephrenis* but not in *X. ramesis* (Table 3). In the progenitor *X. conformis*, mothers fed on the original host produced larger new female imagoes than those fed on the novel host (Tukey's test, t -ratio = 2.90, $P = 0.05$), whereas the descendant fleas of this species produced female offspring of similar size independently of the host they exploited (Tukey's test, t -ratio = -1.11, $P = 0.99$) (Fig. 3A). Regarding between-generation within-host comparison, female offspring of the descendants were larger than those of the progenitors if both fed on the novel host (Tukey's test, t -ratio = -3.37, $P = 0.009$)

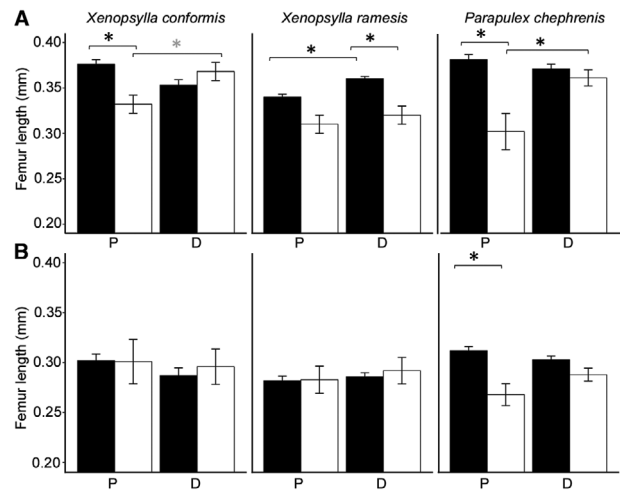


Fig 3 Estimated marginal mean ($\pm$ SE) length of femur of female (A) and male (B) new imagoes produced by *Xenopsylla conformis*, *Xenopsylla ramesis*, and *Parapulex chephrenis* after feeding on either an original host (black columns) or a novel host (white columns). The original host is the host, which fleas of a given species infest naturally. Laboratory colonies of fleas were bred on their original hosts (*X. conformis* and *X. ramesis* on *Meriones crassus*, *P. chephrenis* on *Acomys cahirinus*) and then maintained on the novel hosts (*X. conformis* and *X. ramesis* on *Acomys russatus*, *P. chephrenis* on *M. crassus*). New imagoes were produced by progenitors of each line (parent generation; P) and their descendants of 23rd generation (D). Asterisks denote significant differences. Gray asterisk denotes marginal significance ($P = 0.06$).

and tended to be smaller than those of the progenitors if both fed on the original host (Tukey's test, t -ratio = 2.83, $P = 0.06$) (Fig. 3A).

Descendant, but not progenitor, *X. ramesis* produced larger female offspring if they fed on the original than on the novel host (Fig. 3A) (Tukey's tests; t -ratio = 3.16, $P = 0.04$ for descendants and t -ratio = 2.45, $P = 0.10$ for progenitors). Furthermore, the descendant *X. ramesis* exploiting the original host produced larger female offspring than their progenitor counterparts (Tukey's test, t -ratio = -3.45, $P = 0.03$), whereas female offspring of similar size were born from both progenitors and descendants exploiting the novel host (Tukey's test, t -ratio = -0.70, $P = 0.99$) (Fig. 3A).

In progenitor *P. chephrenis*, female offspring were significantly larger when their mothers fed on the original than on the novel host (Tukey's test, t -ratio = 3.80, $P = 0.02$), whereas female offspring of descendent mothers fed on different hosts did not differ in size (Tukey's test, t -ratio = 1.04, $P = 0.99$) (Fig. 3A). In other words, female offspring size did not differ between the progenitor

Table 2 Summary of the generalized linear mixed-effects models of the effect of host identity (HI) and generation (GEN) and their interaction on the size of eggs produced by *Xenopsylla conformis* (Xc), *Xenopsylla ramesis* (Xr), and *Parapulex chephrenis* (Pc). Fleas were bred on their original hosts (*X. conformis* and *X. ramesis* on *Meriones crassus*, *P. chephrenis* on *Acomys cahirinus*) and then maintained on the novel hosts (*X. conformis* and *X. ramesis* on *Acomys russatus*, *P. chephrenis* on *M. crassus*). Eggs produced after feeding on either the original or the novel host were measured for progenitors of each line (parent generation) and their descendants of 23rd generation (independent variable GEN). Oviposition rate (number of eggs produced per female; EF) was included as an additional explanatory variable in each model. Random factor in all models was a group of parent females (see text for details). Reference levels of independent variables were the original host for HI and parent generation for GEN. R^2c and R^2m are conditional and marginal r^2 , respectively. Significant P -value is in bold.

Flea	Explanatory variable	Coefficient $\pm$ SE	t	P	R^2c	R^2m
Xc	HI	-1.25 ± 0.71	-1.77	0.12	0.14	0.07
	GEN	0.35 ± 0.38	0.94	0.39		
	HI $\times$ GEN	0.27 ± 0.83	0.33	0.75		
	EF	-0.02 ± 0.31	-0.08	0.94		
Xr	HI	0.89 ± 1.11	0.80	0.45	0.14	0.02
	GEN	0.22 ± 0.49	0.46	0.66		
	HI $\times$ GEN	-1.73 ± 0.87	0.06	0.06		
	EF	0.23 ± 0.31	0.72	0.49		
Pc	HI	0.19 ± 0.43	0.45	0.65	0.04	0.03
	GEN	0.22 ± 0.14	0.15	0.15		
	HI $\times$ GEN	-0.15 ± 0.43	-0.35	0.72		
	EF	0.26 ± 0.10	2.67	0.01		

and descendant fleas exploiting the original host (Tukey's test, t -ratio = 1.27, $P = 0.86$). However, the descendant *P. chephrenis* produced significantly larger female new imagoes than their progenitors when both fed on the novel host (Tukey's test, t -ratio = -2.92 , $P = 0.02$) (Fig. 3A).

The effects of larval density and duration of development on female offspring size were significant and consistently negative in all fleas (Table 3). In other words, (i) higher larval density caused female offspring to be smaller and (ii) smaller females took longer to develop. In all LMEs, conditional R^2 was approximately 1.5 times greater than marginal R^2 indicating importance of the random effect (random variation in responses between flea groups).

Both progenitor and descendant *Xenopsylla* produced male offspring of similar size independently of host identity (Table 3; Tukey's tests within a generation between host species, t -ratio = -0.53 to 0.07 , $P > 0.99$ for all; Tukey's tests within a host species between progenitors and descendants, t -ratio = -1.00 to 1.42 , $P > 0.70$ for all) (Fig. 3B). In *P. chephrenis*, the size of male offspring was affected by host identity differently in the progenitor and descendant fleas. Larger new male imagoes were born from the progenitor mothers fed on the original than

on the novel host (Tukey's test, t -ratio = 3.70, $P = 0.003$) and no size difference was found between male offspring of the descendant fleas independently of the host they fed on (Tukey's test, t -ratio = 2.06, $P = 0.19$) (Fig. 3B). Nevertheless, progenitor and descendant *P. chephrenis* exploiting either the original or the novel host produced similar-sized male offspring (Tukey's tests, t -ratio = 1.53 and t -ratio = -1.78 , $P = 0.34$ for both).

The effect of larval density on the size of male offspring was significant in *P. chephrenis* only (Table 3). The direction of this effect was negative suggesting smaller males emerging at higher larval density. Similar to female offspring, male offspring of all three species demonstrated negative relationship between duration of their development and size (Table 3). Conditional R^2 of LMEs for male offspring size was substantially greater than marginal R^2 except for *X. ramesis* in which difference between these coefficients of determination was less pronounced (Table 3).

Discussion

The results of this study demonstrated that the patterns of performance of the progenitors and descendants of the

Table 3 Summary of the generalized linear mixed-effects models of the effect of host identity (HI) and generation (GEN) and their interaction on size of new imagoes (F, females and M, males) produced by *Xenopsylla conformis* (Xc), *Xenopsylla ramesis* (Xr), and *Parapulex chephrenis* (Pc). Fleas were bred on their original hosts (*X. conformis* and *X. ramesis* on *Meriones crassus*, *P. chephrenis* on *Acomys cahirinus*) and then maintained on the novel hosts (*X. conformis* and *X. ramesis* on *Acomys russatus*, *P. chephrenis* on *M. crassus*). Size of new imagoes produced after feeding on either the original or the novel host was estimated for progenitors of each line (parent degeneration) and their descendants of 23rd generation (independent variable GEN). Total number of eggs produced by a group of parent females (EN) and duration of development of each new imago (DD) were included in each model as additional explanatory variables. Random factor in all models was the individual number of a group of parent females (see text for details). Sex, sex of a new imago. Reference levels of independent variables were the original host for HI and parent generation for GEN. R^2c and R^2m are conditional and marginal r^2 , respectively. Significant P -values are in bold.

Flea	Sex	Explanatory variable	Coefficient $\pm$ SE	t	P	R^2c	R^2m
Xc	F	HI	-0.04 ± 0.01	-2.90	0.01	0.71	0.49
		GEN	-0.02 ± 0.008	-2.82	0.01		
		HI $\times$ GEN	0.05 ± 0.01	4.41	0.0001		
		EN	-0.002 ± 0.0007	-3.35	0.005		
		DD	-0.003 ± 0.0005	-5.56	0.0001		
	M	HI	-0.002 ± 0.02	-0.07	0.92	0.52	0.18
		GEN	-0.01 ± 0.01	-1.42	0.19		
		HI $\times$ GEN	0.01 ± 0.02	0.50	0.62		
		EN	-0.001 ± 0.001	-1.40	0.20		
		DD	-0.002 ± 0.001	-2.06	0.04		
Xr	F	HI	-0.03 ± 0.001	-2.46	0.02	0.71	0.49
		GEN	0.02 ± 0.005	3.48	0.007		
		HI $\times$ GEN	-0.01 ± 0.01	-0.96	0.34		
		EN	-0.003 ± 0.0005	-6.76	<0.001		
		DD	-0.003 ± 0.0005	-6.17	<0.001		
	M	HI	0.001 ± 0.01	0.07	0.94	0.59	0.42
		GEN	0.004 ± 0.006	0.69	0.50		
		HI $\times$ GEN	0.004 ± 0.01	0.39	0.70		
		EN	-0.001 ± 0.0006	-1.57	0.15		
		DD	-0.002 ± 0.0003	3.84	<0.001		
Pc	F	HI	-0.07 ± 0.02	-3.80	<0.001	0.68	0.38
		GEN	-0.01 ± 0.008	-1.27	0.21		
		HI $\times$ GEN	0.06 ± 0.02	3.18	<0.001		
		EN	-0.002 ± 0.0005	-4.82	<0.001		
		DD	-0.0008 ± 0.003^s	-2.39	0.02		
	M	HI	-0.08 ± 0.002	-3.80	<0.001	0.69	0.38
		GEN	-0.01 ± 0.007	-1.27	0.14		
		HI $\times$ GEN	0.07 ± 0.02	3.18	0.03		
		EN	-0.002 ± 0.005	-4.82	<0.001		
		DD	-0.001 ± 0.0003	-2.39	<0.001		

experimental lines on the novel host as compared to the original host differed between (i) performance variables and (ii) the three flea species. Specifically, egg and offspring production did not increase on the novel host and did not decrease on the original host even after 23 generations spent on the novel host. The same was true for egg size and probability of an egg to survive until larval

hatching. On the contrary, the descendants of the experimental lines of the host specialist (*P. chephrenis*) and one of the host generalists (*X. conformis*) species produced larger female, but not male, offspring as compared with the progenitors of these lines when feeding on the novel host. Importantly, when feeding on the original host, descendant *X. conformis* tended to produce smaller female

imagoes than their progenitors, whereas there was no decrease in female offspring size between progenitor and descendant *P. chephrenis*. In other words, at least one of the host generalists demonstrated some tendency to become a host specialist (i.e., increased performance on the novel host with a concomitant decreased performance on the original host), whereas the host specialist demonstrated some tendency to become a host generalist (i.e., increased performance on the novel host without a concomitant decreased performance on the original host).

Transition from low to high host specificity and vice versa

A decrease in the ability to exploit the original host while adapting to a novel host was found in the generalist but not the specialist flea, whereas both generalist and specialist demonstrated increased performance on the novel host. This suggests that the effects of (i) “familiarity” with multiple hosts (Joshi & Thompson, 1995) and (ii) “preserved plasticity in host use” (Agosta & Klemens, 2009) may determine whether a trade-off in performance on the novel versus the original host arises but are not associated with a probability to adapt to the novel host and thus do not determine the direction of transition from either generalist to specialist or vice versa. In other words, both directions are probable. In fact, phylogenetic patterns of the level of host specificity of fleas demonstrated, although in an indirect way, that host specificity may have tended to decrease in some flea lineages due to the benefits of exploiting a wide range of host species, whereas in other lineages no particular direction of host specificity evolution was apparent (Poulin et al., 2006). Fox and Morrow (1981) argued that the degree of resource specialization is a local phenomenon developed in a response to local conditions rather than a species-specific attribute. However, this concept may hold for phytophagous insects, but it does not hold for fleas because the level of host specificity in these insects represents true species character and varies significantly more among species than among populations of the same species (Krasnov et al., 2004a). Furthermore, the degree of host-specificity in fleas appears to be scale-invariant (Krasnov et al., 2008). In other words, a flea exploiting a narrow host spectrum in a particular location also exploits a narrow spectrum of principal hosts across its geographic range and is unable to change its principal host species across regions, whereas a locally host-opportunistic flea appears also to be host-opportunistic on a larger spatial scale and often change its principal hosts across regions.

A few earlier experimental studies testing for the reproductive trade-offs in fleas maintained over generations on different hosts provided contradictory results. For example, Arbiv et al. (2012) found that fleas *X. ramesis* maintained on a particular host species for a few generations substantially decreased their reproductive performance when transferred to an alternative host species, whereas they generally did not increase their performance on their maintenance host. This pattern did not depend on whether the maintenance and the alternative hosts were the typical and the atypical hosts for a flea, respectively, or vice versa. On the contrary, Stavtseva et al. (2021) found no indication of a loss of reproductive performance on the original host while adapting to a novel host in *X. ramesis* and *Synosternus cleopatrae* (Rothschild, 1903). However, these studies dealt with closely related hosts that were likely similar in the quality of resource they provide (blood), pattern of its acquisition (skin thickness, hair density) and in their behavioral and/or immunological defenses against fleas. The patterns of flea response to distantly related hosts could be different because differences in performance in different environments are expected to be more pronounced between populations from more strongly divergent environments (Hereford & Winn, 2008). Indeed, hosts used in this study belong to different subfamilies and thus likely substantially differ in their blood composition, skin structure and behavior. In this study, patterns found in *X. ramesis* contradicted to those of Arbiv et al. (2012) and supported those of Stavtseva et al. (2021). Indeed, *X. ramesis* did not increase performance on the novel host and did not decrease it on the original host. However, production of larger female offspring by descendant *X. ramesis* when they returned on their original host suggests some compensatory mechanism after being forced to exploit resources of presumably low quality during multiple generations. This mechanism undoubtedly warrants further investigation.

Benefits of larger female offspring size

Although *X. conformis* and *P. chephrenis* adapting to the novel hosts did not increase their fitness directly (neither egg number nor egg survival changed), they undoubtedly benefited by producing larger female offspring. Furthermore, higher quality (i.e., larger size) of female new imagoes born from the descendant mothers after feeding on novel hosts was not a result of an often-observed trade-off between quantity and quality of offspring (Lack, 1947; Smith & Fretwell, 1974) because the numbers of eggs and new imagoes did not differ between the progenitors and descendants exploiting the novel host. Further-

more, the trade-off between the offspring quantity and quality usually arises due to resource limitation (Stearns, 1992), so that mothers have to make fitness-related decision whether to invest more in the number of their offspring or in their size. Fleas, like many other parasites, often do not face resource limitation and thus no trade-off between the number and the size of offspring in these insects was found (Kiefer *et al.*, 2016; this study). Nevertheless, Khokhlova *et al.* (2014) reported a negative relationship between the number of eggs and the size of the offspring in *X. ramesis* and *P. chephrenis*. This inconsistency might be associated with the fact that the size of a new imago in holometabolous insects can be affected not only by the size of an egg (as well as environmental conditions such as ambient temperature) but also by larval feeding. This is supported by negative relationship between the number of eggs produced by group of female fleas (a proxy for larval density) and the size of female (for both *Xenopsylla*) and both female and male (for *P. chephrenis*) offspring suggesting some level of competition among larvae. This competition could be both exploitative and interference. In the former case, larger larvae are more efficient foragers, assimilate resources faster than smaller larvae, and outcompete them (Fox, 1994; Khokhlova *et al.*, 2010). In the latter case, larger larvae cannibalize smaller larvae (Lawrence & Foil, 2002). Higher foraging efficiency of larvae developing into larger new imagoes is indirectly supported by a negative relationship between the size of a new imago and duration of development (see also Kiefer *et al.*, 2016). In addition, the experimental designs of Khokhlova *et al.* (2014) and this study were different. In particular, the occurrence of the trade-off between quantity and quality of flea offspring in Khokhlova *et al.* (2014) was found when mother fleas exploited different host species, whereas in this study they fed only on the same host species.

Kiefer *et al.* (2016) found that female body size in fleas was related to postoviposition survival with larger females surviving longer after oviposition than smaller females. Longer survival allows a larger female to have more opportunities for future reproduction resulting in higher potential lifetime fecundity than a smaller female (Berger *et al.*, 2008). This suggests that fleas adapting to a novel host and producing larger female offspring benefit from increased lifetime, rather than immediate, fecundity of these offspring. Nevertheless, neither adaptation to a novel host (in *X. conformis* and *P. chephrenis*) nor the decreased success in exploitation of an original host (in *X. conformis*) was manifested via the size of the male offspring. This might be explained by stronger selection pressure on body size in females than in males because larger body size in females likely results in higher

life-time fecundity (Henry *et al.*, 2009; Chown & Gaston, 2010; Kiefer *et al.*, 2016), whereas larger males do not always demonstrate higher fitness than smaller males (Kelly *et al.*, 2008). Instead, smaller males may have an advantage over larger males in mate competition due to higher mobility (Kelly *et al.*, 2008).

Differential response to a novel host between two generalists

In our study, a tendency to become a host specialist after multiple generations on a novel host was found in only one of the two generalist species. This difference might be associated with a differential host-selection strategy (Krasnov *et al.*, 2003c, 2004b). Analyses of field data on flea distribution on different hosts (Krasnov *et al.*, 2003c), which were supported later by laboratory experiments (Krasnov *et al.*, 2004b), showed that *X. conformis* is characterized by sharp host selectivity even at low density. This is because of a higher fitness reward for individuals exploiting a particular host species than that for individuals exploiting other host species. In contrast, the natural distribution of *X. ramesis* indicated random host selection even at low density and fleas of this species did not demonstrate any direct reproductive advantage when feeding on a particular host.

In conclusion, the probability of a generalist to specialist transition may be context-dependent and vary between flea species. Nevertheless, it is difficult to give a clear-cut answer about between-species variation in the probability of specialist to generalist transition because no laboratory colony is available for any host-specific flea other than *P. chephrenis*; therefore, there is no opportunity to collect comparative data.

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Disclosure

The authors have no conflict of interest to declare.

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